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ARTICLE

Evaluation of the efficiency of urban solid waste management in Brazil by data envelopment analysis and possible variables of influence

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Abstract

This article has the general objective of estimating the efficiency of urban solid waste management in 940 Brazilian municipalities through Data Envelopment Analysis (DEA) technique and has specific objectives: (i) to estimate efficiency scores; (ii) to compare the performance between different groups of municipalities; and (iii) to analyze the profile of efficient municipalities from the perspective of the guidelines of Law 12,305/2010 and socio-economic and environmental indicators. The technique used was DEA with output-oriented and variable scale to return modeling. The results showed higher efficiency scores in the municipalities with populations above 500,000 inhabitants. The score variation ranged from 0.5 (municipalities with populations <10,000 inhabitants) to 0.9 (municipalities with more than 500,000 inhabitants). Of the sample set, only 12.34% of the municipalities were considered efficient, and when analyzing the efficient group, it was found that adherence to legislation was not a major factor in achieving efficiency.

Keywords Urban solid waste management · Data envelopment analysis · Efficiency · Brazil · Municipalities

Introduction

Besides debates on global warming, biological diversity, and pollution, the issue of adequacy of waste management has also been discussed in several groups [1], especially because, annually, 2 billion tons of urban solid waste are generated in the world and projections for the year 2050 reveal increases of 19% in developed countries and 40% in developing countries [2]. Considering that at least one-third of the waste produced in the world is inadequately managed [2], solutions based on the adequacy of the final disposal system, to the detriment of policies aimed at reducing or reusing waste, are no enough to contain the socio-environmental impacts of inadequate management. Likewise, the lack of strategies to reduce waste at its source has made the impact of waste policies insignificant [3].

The global pressures for the adequacy of waste management have made high-income countries the pioneers in the reformulation of normative instruments of the Urban Solid Waste Management (USWM) [4], citing as an example, the member countries of the European Union (EU) with the respective years of approval of the legislation (France in 1957, Germany in 1986, Spain in 1998, and Portugal in 1997). In the United States, the law was approved in 1965 and in Japan in 1970 [5].

The challenges for adequacy in the USWM are greater for middle- and low-income countries [6], as they are still trying to overcome primary problems of the USWM, such as the collection and removal of its waste [7]. In addition, the scarcity of financial resources, the lack of qualified labor, and low social involvement are significant obstacles to the adequacy of management in these countries. This context makes it imperative to develop research with a focus on waste management in medium- and low-income countries.

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Classified as a middle-income country, Brazil has a population of 203,080,756 inhabitants [8] distributed in 5,570 municipalities, and according to data presented by Kaza et al. [2], the country that in 2015 generated about 79 million tons of urban solid waste occupies the fourth position in the world in generation of urban solid waste, only behind: the United States with 258 million tons, China with 220 million tons, and India with 168 million tons of municipal solid waste (MSW) generated [2].

The regulation of USWM in Brazil only took place in 2010 through the National Waste Policy [9], which is why it is considered long-overdue in relation to developed countries [10]. The difficulties in implementing the law are notorious [11]; in addition, the country lacks political instruments aimed at USWM processes and a more efficient management model [12].

Some contexts demonstrate flaws in the implementation of Law [9], given that in 2019, 40% of the urban solid waste generated in the country was still directed to final disposal systems prohibited by law [13], and after 3 years, the decrease of this reality was not significant, as 39% of urban solid waste, or the equivalent of 29.7 million tons, continued to be directed to inappropriate locations (dumps or controlled landfills) [14].

Another factor to be considered is that although collection coverage in Brazil has reached 89.9% of the population, 22 million inhabitants still do not have access to this service, of which 19 million live in rural areas [15]. This context reveals a significant number of inhabitants marginalized from waste collection [16] and a great disparity in this process between urban and rural areas, with a downward deviation in rural and peri-urban areas [17]. Discrepancies in collection coverage between the regions of the country are also a reality, with losses for the less developed regions, such as the North and Northeast, where the lack of collection corresponds to 21% and 17.6%, respectively, while the more economically developed regions, such as the Southeast, South, and Midwest, have the lowest percentages, 4.2%, 8.4%, and 9.1%, respectively [15]. With regard to selective collection initiatives, they are present in only 32% of the municipalities, ranging from 10% in the Northeast region to 56.9% in the South region [15].

Although more than a decade has gone by since the approval of the National Solid Waste Policy [9], most Brazilian municipalities have deficiencies in various waste management processes, such as collection, final disposal, and reuse.

Among other factors, it should be highlighted that the USWM adequacy process depends directly on both the evaluation of political instruments and their applicability and the analyses carried out to measure the efficiency of USWM. This is because the results of these analyses are capable of portraying the reality of the waste sector in a given locality and suggesting more assertive directions in the adequacy process.

Referring to Brazil, there are no scientific bases indexed in any research that measures the efficiency of the USWM and establishes relations with the directives of legislation and/or socio-economic indicators, a context on which this context greatly expands the relevance of the development of research in nature.

In this sense, this article has the general objective of estimating the efficiency of the USWM of 940 Brazilian municipalities through Data Envelopment Analysis technique (DEA) and has specific objectives: (i) to estimate efficiency scores; (ii) to compare the performance between different groups of municipalities and (iii) to analyze the profile of efficient municipalities from the perspective of the guidelines of Law [9] and the socio-economic and environmental indicators.

With regard to methodological aspects, the research included the following protocols: (1) classification of municipalities into 5 groups based on population range (<10,000 inhabitants; between 10,000 and 30,000; between 30,000 and 100,000; between 100,000 and 500,000, and over 500,000 inhabitants); (2) definition of variables for generating efficiency scores: Inputs (Number of Employees in collection; Number of Vehicles used in collection and Expenses with collection) and Outputs: (Collected Volume and Served Population); (3) generation of efficiency scores through the DEA technique oriented to the outputs with variable scale to return modeling; and (4) analysis of the profile of efficient municipalities to perceive possible influences of the following external variables: Municipal Waste Management Plan, Final Disposal System, Social Control, Regulatory Body, Selective Collection, Cooperatives/Associations (guidelines of Law [9]); Population, Demographic Density, Gross Domestic Product, Municipal Human Development Index (MHDI), Average Monthly Salaries, and Percentage of Adequate Sanitary Sewage (socio-economic and environmental indicators).

The theoretical basis of this research considered classic scientific references and more recent publications extracted mainly from the Scopus database. International and national reports, as well as the Brazilian Legislation of the USWM were also used to collect data and information for this research.

Literature review

Previous scientific research on the evaluation of USWM efficiency through DEA

The analysis focused on the efficiency of the management of urban solid waste is as essential as it is necessary to present diagnoses and help to face weaknesses, especially in low- and middle-income countries, the vast majority of which have their waste managed by the public sector.

The non-parametric technique called DEA has proven to be suitable for analyzing the efficiency of Decision-Making Units (DMUs) [18, 19] and the main characteristics of this technique include: the non-requirement of restrictive assumptions about the data generation process [20] and the waiver of the complete specification of the functional production format [21, 22].

Introduced by Charnes et al. [23], DEA aims to support the decision-making process by measuring the efficiency of resource use, in whatever combinations are present and considering the technologies available in the production process [23]. With the use of linear mathematical programming, it is possible to obtain ex post-evaluations related to the relative efficiency of management achievements, regardless of how they may have been planned or executed [24].

The initial applications of DEA focused on evaluating non-profit units inherent to public programs [23]. Therefore, the public sector was considered the engine for the propagation of this technique, having reached both new possibilities of empirical evaluation of its efficiencyas well as opportunities to better manage expenses [20].

When estimating the efficiency of a set of companies, herein called DMUs, DEA does not assume any prior assumptions in the production model and manages to show the best performances [22, 25–28]. Through the conversion of multiple inputs and multiple outputs of the DMUs, efficiency indices are available on a scale that varies from zero to one, where when a DMU reaches the highest score on the scale corresponding to 1 unit, it is considered efficient becoming a benchmark for inefficient DMUs [26].

Although applied research in Brazil to measure the efficiency of USWM through DEA is rare, the use of this method in analyses of this nature is common in the scientific literature. Examples applied in Brazil that stand out are de Oliveira et al. [29] who measured the level of efficiency of USWM in Brazilian capitals during the period from 2002 to 2016, highlighting the Southern region as the one with the best performance, and Fidelis and Colmenero [30] who evaluated the managerial performance of seven recycling Brazilian cooperatives.

In other locations, relevant studies analyzing the efficiency of USWM deserve to be mentioned, such as Yang et al. [19], who evaluated 33 Chinese cities to measure the efficiency in the routes of implementing USWM policies, noting the low efficiency of the Chinese USWM and regional differences in performance that suggest the application of policies appropriate to the characteristics of each region.

In Italy, their search which stands out is by lo Storto [31], who, when evaluating the effect of explanatory variables on the performance of MSW management in 258 municipalities in the Apulia region (southern Italy), found that the reduction of selective collection implies a reduction of USWM efficiency; in addition, the improvement of MSW services involves reducing the average amount of waste generated. Sarra et al. [32] in turn analyzed the efficiency of the USWM of 289 Italian municipalities in the period 2011–2013, having found a negative effect on the efficiency of the exogenous variables: municipality area, elevation above sea level, and spatial dispersion index of the population, and Spallini et al. [33], having measured the efficiency scores of 103 Italian municipalities, identified connections between the scores and the type of management adopted.

The Japanese context was observed by Ichinose et al. [34] who after analyzing 47 city halls identified greater efficiency in those whose USWM services were performed by the private sector and less efficiency in those with remotely inhabited islands, reinforcing the influence of geographic factors on performance. Chen [35] presents relevant contributions to the waste area by assessing the integrated efficiency of USWM in 23 municipalities in Taiwan (China), emphasizing three processes (collection, classification, and generation). In addition to measuring the separate efficiency of each process, the author compares the performance of urban and rural areas, observing better classification performance for the urban area and better performance in the generation and collection processes for the rural area.

To present Spanish performance, Benito-Lopez et al. [20] estimated the determinants of the efficiency of the USWM of 1072 Spanish municipalities and found a significant and positive relationship between the efficiency and external variables: per capita income, comparative index of importance of tourism, and economic activity. With a similar approach, Simões et al. [26] calculated the influence of the operating environment on the efficiency of USWM services provided by 29 concessionaires (public and private) serving the mainland area of Portugal. These authors found that private

concessionaires had higher efficiency scores and a significantly positive relationship between GDP and efficiency, and a negative relationship between efficiency and the variables population density and distance from the landfill.

More expansive investigations with the use of the DEA method were also identified in the literature, such as in the analyses of the environmental and circular economy models of the USWM of 27 countries of the EU, as well as in the consequent ranking of the best performers [36], and last but not least, in the assessment of environmental efficiency of 22 countries- Member States of the EU, in the years 2005, 2010, and 2015 [6].

The regular frequency of the use of the DEA technique to assess the efficiency of USWM in different locations around the world, according to the scientific studies cited in this section, reveals the adequacy of this technique and supports the methodological protocols of this research that are detailed in the subsequent section.

Methodology

Model and applied variables

The results of the DEA technique can assume constant returns to scale (CCR), an approach proposed by Charnes, Cooper, and Rhodes in 1978, or variable returns to scale (BCC), proposed in 1984 by Banker, Charnes, and Cooper. In the CCR model, it is assumed that the increase in resources of a DMU will cause proportional increases in the products, and in the BCC, proportionality between resources and products is not assumed, assuming increasing returns to scale, constant returns, and decreasing returns to scale [37]. This paper employs the basic BCC DEA model, which considers variable returns to scale. The option for the BCC model follows the recommendation for analyzing DMUs that operate in a scenario of imperfect competition due to resource constraints [38].

The DEA model allows the data to be executed, oriented to input or output, and the signals provided to reach optimal production levels are related to the choice of orientation to be applied. Thus, when oriented to input, the manager can reduce the inputs, keeping the outputs fixed, and when oriented toward output, the manager can expand the outputs, keeping the inputs fixed.

The output orientation was the one defined for the model of this research, since the waste management service in Brazil is densely offered by the public sector, which frequently operates, with scarce resources and faces the challenge of increasing efficiency, and maintaining fixed resources. The choice of model orientation should be based on the quantities (inputs/outputs) over which managers have the most control, so in contexts where DMUs are given a fixed amount of resources to produce as much as possible, the output orientation can be the most adequate [38]. Moreover, it is not the objective of municipalities to reduce their outputs, but to expand their inputs [39].

The output-oriented BCC model can be represented by Eqs. 1 to 3 according to Lima et al. [39]

$$E = Min \sum_{j=1}^{n} v_j.x_{j0} - w \tag{1}$$

subject to

$$\sum_{i=1}^{m} u_i.y_{i0} = 1 \tag{2}$$

$$\sum_{i=1}^{m} u_i.y_{ik} - \sum_{j=1}^{n} v_j.x_{jk} + w \leq 0, \quad tok = 1, 2, \ldots h \tag{3}$$

w no signal restriction

$v_j, u_i \geq 0$

$j = 1, \ldots n;\ i = 1, \ldots m,$

where

x_{jk} is the amount of the input j of the DMUk;

y_{ik} is the amount of the output i of the DMUk;

x_{j0} is the amount of the input j of the DMU under analysis;

y_{i0} is the amount of the output i of the DMU under analysis;

v_j is the weight of the input j for the DMU under analysis;

u_i is the weight of the output i for theDMU under analysis;

w is the scale factor;

m is the number of analyzed outputs;

n is the number of analyzed inputs; and

h is the number of DMU.

Regarding the definition of variables to measure efficiency, two criteria were established: availability of data in the base of the National Sanitation Information System (SNIS) in the year 2019 [40] and scientific relevance. Thus, the following variables were defined: 3 Inputs (Expenses with the collection of urban solid waste, Number of employees in the collection and Number of vehicles used in the collection) and 2 outputs (Volume collected and Population served).

Previous studies support the choice of the variables used in this research, namely, the expense variable was used by

Simões et al. [26], to assess the efficiency of 178 USWM service providers in Portugal by Amaral et al. [41], to estimate the efficiency of collection services provided to 196 Portuguese municipalities by Simões et al. [27], to assess the impacts of local policies on the efficiency of waste treatment and final disposal in 299 Belgian municipalities by De Jaeger et al. [42], and to measure the performance of waste collection and processing in 293 Belgian municipalities by Rogge and De Jaeger [21].

The use of the Number of employees and Number of vehicles variables was made by Sánchez [43] when evaluating collection performance in 34 Spanish municipalities, as well as by Ichinose et al. [34] and Fidelis and Colmenero [30].

In this research, the waste collected and population served were considered as the desired output of the model, as they can be a proxy for the efficiency of the collection system. Although in other research, as in Lombardi et al. [44], the collected waste is an undesirable output, as this means that the population is recycling less, in developing countries like Brazil, the stage of development of the waste makes the efficiency measure more meaningful based on the amount collected rather than the amount that is not recycled. In addition, the large population dispersion and inequality in access to the collection service in Brazil mean that the larger the population served, the greater the efficiency of the system should be.

Some noteworthy references, who also considered the variable volume collected, are Expósito and Velasco [25], who evaluated the performance of the recycling market in 17 Spanish regions, and Yeh et al. [28] who analyzed the performance of recycling systems in Taiwan (China). In addition, notable research examples that used the population served variable studies of a similar nature are Chen [35], and Fidelis and Colmenero [30].

To achieve one of the objectives of this research, variables external to the USWM process were included in the analysis model to present the profile of efficient DMUs and discuss possible relationships between efficiency and these variables.

Thus, the external variables defined were 6 guidelines of Law [9] (Municipal Management Plan, Social Control, Regulatory Body, Appropriate Final Disposal System, Selective Collection, and Recycling Cooperatives) and 6 socio-economic and environmental indicators (Population Demographic Density Product Gross Domestic, MHDI, Average Monthly Salary, and Percentage of Adequate Sanitary Sewage).

The criteria for selecting the guidelines of law 12,305/2010 took into account the availability of data in the SNIS base [40], and to select the indicators, the scientific relevance and the importance attributed by the authors to the indicator were considered. Indicator data were extracted from the Brazilian Institute of Geography and Statistics. The 6 guidelines of Law 12,305/2010 used in this research were also used by das Mercês Costa et al. [45] to analyze the USWM profile of 786 Brazilian municipalities, noting that only 5.6% of municipalities' evaluations adhered to the guidelines.

Some of the aforementioned indicators were used by other researchers as exogenous variables to verify the influence on the efficiency of the USWM through a two-stage evaluation, for example, Simões et al. [27], De Jaeger et al. [42], and Romano and Molinos-Senante [46] used Population, while Benito-López et al. [20], Expósito and Velasco [25], Simões, et al. [27], De Jaeger et al. [42], Simões et al. [26], Romano and Molinos-Senante [46] used Demographic Density. On the other hand, Gross Domestic Product Per Capita was used by Simões et al. [27], and the MHDI was used by das Mercês Costa et al. [45]. Although it has not been identified in the literature, the authors considered it pertinent to include the percent of Sanitation Coverage to meet the environmental dimension and the Average Monthly Salary to strengthen the analyses of the social and economic dimensions of the municipalities. The description of the variables is shown in Table 1.

Definition and characterization of the research sample

With 5570 municipalities distributed across 27 Federative Units (FU), Brazil is the fifth largest country in the world in territorial extension, with an area of 8,510,345,538 km^2 [47]. Geographically, the country is divided into 5 regions: North, Northeast, Midwest, Southeast, and South, and has an estimated population of 203,080,756 inhabitants [47].

The empirical analyses involved the municipalities where all the variables defined for the model were available in the SNIS database for the year 2019, therefore, of the 3712 municipalities that are part of that database, it was possible to include 940 municipalities in the sample. Considering this sample, 84,350,822 million inhabitants or equivalent to about 40% of the estimated population in 2019 [48] were involved in this research.

Although the objective of this research is not to carry out regional analyses, to broaden the reader's perception of the sample's characteristics, we gathered, data related to the percentage and numerical distribution of the sample in Table A.1, considering the regions and FU of the country. In summary, the Southeast and South regions of the country are responsible for covering 71% of the sample set, and at federative level, the 3 largest concentrations of the sample are in the FUs of: São Paulo, Rio Grande do Sul and Paraná, and the 3 smallest concentrations were registered for the FUs of: Acre, Amapá, and Alagoas (Table A.1).

Table 1 Variables used in the analysis

No/code	Name	Unit of measurement	Description
Variables for measuring efficiency scores			
COL_EXP (Input)	Expense on collection of USW	BRL	Expenses on the collection of municipal solid waste/year
COL_EMP (Input)	Workers in the collection	No. Person	Number of workers involved with collection
COL_VEHIC (Input)	Vehicles for collection	No. Vehicles	Number of vehicles used for collection
COL_VOL (Output)	Collectedurbanwaste	ton	Volume of waste collected/Year
POP_SERV (Output)	Populationserved	No. Person	Number of people served by the collection service
Socio-economic and environmental indicators			
POP	Population	No. Person	Number of inhabitants in the municipality
DD	Demographicdensity	km^2	Relationship between the population and the area of the territory of the municipality
GDP per capita	Gross Domestic Product—GDP	BRL	Sum of all final goods and services produced in a given region during a given period
MHDI	Municipal human development index	(0–1)	Comparative measure used to classify countries/ municipalities by their degree of human development
AMS	Average monthly salary	Number of Salaries	Average monthly salary of formal workers
%AS	Percentage of adequate sanitation coverage	%	Percentage of Adequate Sanitation Coverage in the Municipality
Guidelines of Law 12,305/2010			
ISWM	Integrated municipal waste management plan	–	Existence of the USWM Municipal Plan Art. No. 26
S_Con	Social control	–	Existence of Social Control of USWM services. Inciso X, Art. No. 6
FDS	Final disposal system	–	Adequacy of final waste disposal systems Inciso II, Art. No. 7, and Art. No. 54
R_Sec	Regulation/Inspection sector	–	Existence of a USWM regulation / inspection sector. Inciso V, Art. No. 8
SC	Selectivecollect	–	Existence of Selective Collection Inciso VIII, Art. No. 6, and Inciso III, Art. No. 8
C/A	Cooperatives/ Associations	–	Existence of cooperatives and/or associations for the reuse or recycling of waste. Inciso XII e XIV, Art. No. 7, and Inciso IV, Art. No. 8

Source: Data from this research, 2022; *BRL* Brazilian Real

To satisfy the proposed analysis, the population range was defined as a classification parameter, since the population reflects both consumption patterns and the volume of MSW generated. According to Díaz-Villavicencio [49], population is of great importance in the efficiency of waste management, and its size influences costs and economies of scale in waste management. It is noteworthy that the inclusion of population size as municipalities' group classification parameter is not in conflict with the variable used as output (population served), since the collection service in Brazil is not offered to the entire population.

To satisfy the proposed analysis, the municipalities were classified into 5 groups using the population range as a parameter. The choice of the population range is due to the fact that this dimension reflects both consumption patterns and the volume of MSW generated. Thus, the following groups were formed: Group 1 Municipalities with up to 10,000 inhabitants; Group 2 Municipalities with a population between 10,001 and 30,000 inhabitants, Group 3 Municipalities with a population between 30,001 and 100,000 inhabitants; Group 4 Municipalities with a population between 100,001 and 500,000 inhabitants, and Group 5 Municipalities with a population above 500,000 inhabitants.

As it allows for greater comparability between DMUs, the groups' analysis to measure the efficiency of the USWM was identified in the evaluation of the efficiency of the collection service in 260 Brazilian municipalities [50], as well as in the investigation into the influence of the type of property in the USWM eco-efficiency in 225 municipalities in the Tuscany region of Italy [46] and in the USWM analysis of 103 Italian cities [33].

Table 2 Statistical description of analysis variables by municipalities' groups

Group			Group 1	Group 2	Group 3	Group 4	Group 5
Variables/measures			< 10,000	10,001–30,000	30,001–100,000	100,001–500,000	> 500,000
INPUTS	COL_EXP (BRL)	Minimum	14,000	20,000	200,000	1,165,081.27	13,067,227.83
		Median	177,333.33	584,696.93	2,314,816.82	9,789,344.61	57,370,937.57
		Maximum	1,399,999.20	5,626,559.23	14,250,898.14	60,164,671.34	1,294,122,055.93
		Standard deviation	155,825.86	692,788.62	2,311,866.91	9, 492, 721.48	262,675,159.28
		Average	210,301.63	752,842.98	2,792,678.97	12,119,573.87	142,148,323.98
		Count (DMUs)	361	304	161	88	26
	COL_EMP(No. Employees)	Minimum	1.00	1.00	2.00	10.00	4.00
		Median	4.00	8.00	22.00	75.50	386.00
		Maximum	37.00	44.00	75.00	315.00	5,400.00
		Standard deviation	2.85	6.46	14.03	60.88	1,364.44
		Average	4.47	9.67	24.82	92.37	881.69
		Count	361	304	161	88	26
	COL_VEHIC (UN)	Minimum	1	1	1	5	19
		Median	3	5	8	22	96
		Maximum	11	36	38	92	679
		Standard deviation	2.09	4.31	5.90	16.32	180.35
		Average	3.17	5.76	9.62	25.96	163.15
		Count	361	304	161	88	26
OUTPUTS	POP_SERV (No Inhabitant)	Minimum	256	3,227	15,505	80,000	488,227
		Median	3,216	12,416	42,958	203,173	1,010,218
		Maximum	9,755	29,229	95,870	491,443	12,141,888
		Standard deviation	2,088.73	5,908.90	19,921.89	110,618.99	2,457,583.54
		Average	3666	13,851	48,073.36	222,175.68	1,845,893.42
		Count	361	304	161	88	26
	VOL_COL (Ton)	Minimum	45.00	160.00	360.00	10,625.90	102,534.90
		Median	810.00	3,972.35	14,897.70	73,914.00	453,264.65
		Maximum	9,930.00	21,040.90	77,912.00	253,086.00	3,843,531.40
		Standard deviation	1,045.10	3,599.02	12,908.47	51,105.20	890,817.63
		Average	1,118.64	4,938.46	18,227.90	81,291.70	731,541.21
		Count	361	304	161	88	26

Source: Data from this research; DMUs, Decision-Making Units

Results

Efficiency scores for municipalities' groups

Before measuring the efficiency of the groups, the statistical description (Table 2) of the variables (inputs/outputs) used to measure the efficiency was presented. Data for these variables were extracted from the 2019 database [40] of the National Sanitation Information System.

As shown in Table 2, it can be seen that the larger the population range is, the lower the concentration of DMUs, therefore, the highest concentrations of DMUS are present in groups 1 and 2, and the lowest are present in groups 4 and 5. The opposite was recorded for the means of the input variables, which increased as the population range increased. For instance, the average collection expense (COL_EXP) ranged from 210,301.63 BRL (group 1) to 142,148,323.98 BRL (group 5); the average number of employees (COL_EMP) ranged from 4.47 (group 1) to 881.69 (group 5); and equally, the average number of vehicles (COL_VEHIC) had an increasing variation, from 3.17 (group 1) to 163.15 (group 5).

The growing trend, as the population range increased, was also identified in the outputs, where the average of the population served (POP_SERV) ranged from 3666 (group 1) to 1,845,893.42 (group 5) and the average of the collected volume (VOL_COL) ranged from 1,118.64 tons (group 1) to 731,541.21 tons (group 5).

Like Spallini et al. [33] and Yang et al. [19], after performing the descriptive analysis, the data were executed in

Table 3 Perceptual proportional efficiency in each municipalities' group

Group	1	2	3	4	5	∑
Population range	< 10,000	10,001 to 30,000	30,001 to 100,000	100,001 to 500,000	> 500,000	
No. DMUs	361	304	161	88	26	940
No. efficient DMUs (by group)	30	31	26	15	14	116
% of Efficient DMUs (by group)	8.31%	10.19%	16.14%	17.04%	53.84%	–
Average group efficiency	0.51	0.65	0.70	0.70	0.91	–

Source: Data from this research, 2022

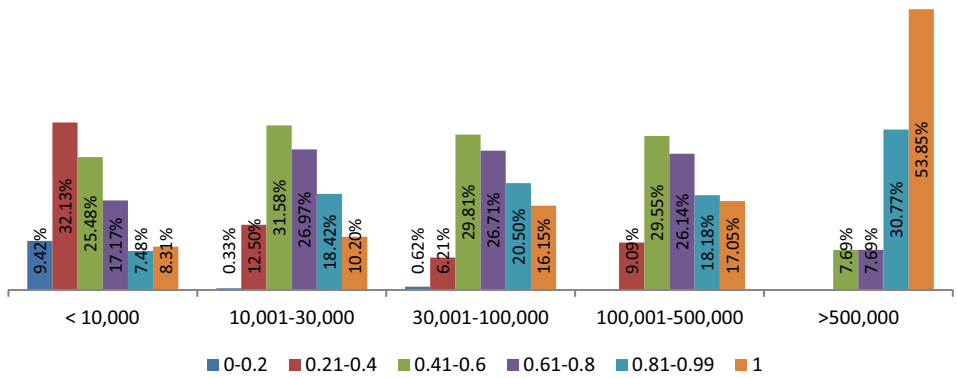

Fig. 1 Municipality groups' performance by efficiency range. Source: Data from this research

DEAP software version 2.1 [38] to measure the efficiency scores of the groups, the results of which are summarized in Table 3.

It can be seen in Table 3 that the larger the population range, the greater the percentage of efficiency, which showing that the most populous Brazilian municipalities (group 5) are the most efficient (53.84%), and the least populous municipalities (group 1) are those with lower efficiency percentages (8.31%). This result converges with Callan and Thomas [51], who concluded that the increase in population reflects a higher level of eco-efficiency in the management of urban waste. The variation in the performance averages increased in relation to the increase in the population range, ranging from 0.508 (group 1) to 0.913 (group 5).

The total number of efficient DMUs considering the 5 groups was 116 (Supplementary material—Table A.2), which represents 12.34% of the sample. This percentage was lower than that of Ferreira et al. [50] who found efficiency in 17% of municipalities.

The performance of the DMUs in each group, considering the efficiency ranges presented in Fig. 1, confirms the best performance for groups 4 and 5. In addition to having the highest percentages in efficiency 17.05% and 53.85%, respectively, there are no DMU, operating in the lowest efficiency range (0–0.2). Similar results were found by Ferreira et al. [50] regarding the presence of a higher percentage of efficiency in municipalities with populations above 500,000 inhabitants. Concerning the DMUs that operate closer to the efficiency frontier with a performance in the range (0.81–0.99), the highest percentages are in group 5 (30.77%) and in group 3 (20.5%).

The percentages of DMUs that operate with an index lower than or equal to 0.6 in groups 1, 2, and 3 were 67.03%, 44.41%, and 36.64%, respectively, strengthening the trend that the lower the population range the lower the percentages of underperforming DMUs. It should be noted that groups 3 and 4 reached a very close average performance, namely, 0.703 and 0.704, respectively, and although the first has a proportional percentage of efficient DMUs (16.15%) lower in relation to the second (17.05%), the percentage of DMUs close to the efficiency frontier (0.81–0.99) is higher for group 3, thus revealing a close relationship in the performance of these two groups.

To verify the profile of efficient DMUs, analyses around the guidelines of Law 12,305/2010 and socio-economic and environmental indicators were carried out and are presented in the following section.

Analysis of the profile of efficient DMUs and discussion of possible influence variables

Considering the 5 groups (12.34%), 116 DMUs achieved efficiency (Table A.2), with the highest incidences being identified in the most developed regions of the country, such

Table 4 Efficient DMUs (Guidelines of Law 12,305/2010 and the socio-economic and environmental indicators)

Code/variable	Variable	Efficient DMUs
Guidelines of Law 12,305/2010		
ISWM Plan	Presence of the ISWM plan	57.76%
SC	Presence of selective collection process	60.00%
C/A	Presence of the cooperatives and/or associations of waste pickers	51.72%
FDS	Adequate final disposal system	56.03%
R_Sector	Presence of the USWM regulatory body	28.45%
S_Con	Social management control initiatives	31.09%
Socio-economic and environmental indicators		
MHDI	Municipal human development index	65.00% have MHDI above 0.7 (high or very high)
DD	Average demographic density	685.68 inhabitant/km^2
AS	Adequate sanitation coverage	61.00%
GDP per capita	GDP per capita above the national average	33.00%
AMS	Average monthly wages paid to formal workers	2.40 salary/month

Source: Data from this research; ISWM, Integrated solid waste management

as the Southeast and South. The Federative Units of São Paulo-SP, Minas Gerais-MG, and Santa Catarina-SC stood out as those with the highest number of efficient DMUs.

Table 4 shows the behavior of efficient DMUs considering the guidelines of Law 12,305/2010 and the socio-economic and environmental indicators. The analysis of the legislation's guidelines did not point to homogeneous results, which suggests that the adequacy to the legislation is preponderant for the achievement of efficiency. There is a polarization in the set of efficient DMUs, with the presence of 2 subgroups, whose percentages of adherence or not to Law 12,305/2010 are around 40% to 60% for most guidelines. The exception in this polarization was for the guidelines with the presence of the regulatory body for management and social control, where the results indicated that the percentage of non-adherence was above 29% for the first guideline and 32% for the second. Despite the legislation not determining the scope of efficiency of the analyzed DMUs, the results of this analysis raise questions about the failure to implement the National Policy in the municipalities.

With regard to data from socio-economic and environmental indicators**,** the Table 4 reveals that the greatest highlight was identified for the MHDI variable, as there were none in the group of 116 DMUs with a low MHDI, that is, with an index lower than 0.55. About 35% have a medium MHDI (0.55 to 0.69), and 65% have a high (0.70 to 0.79) or very high (above 0.80) MHDI. These data suggest that municipalities that invest in the dimensions of health, education, and income (HDI) tend to be more efficient.

For population density, the general average of the group was 685.68 inhabitants/km^2 and the variation ranged from 1.04 in *Alcinópolis-MS* to 9,063.58 in *Olinda-PE* (Table A.2). There was no homogeneity in the group regarding the percentage of adequate sanitary sewage, since there are municipalities in the group such as *Alcinópolis-MS* that only have 0.4% of adequate coverage and others such as *Pontal-SP* with 99.4% coverage. Despite this high variation, it should be mentioned that more than half of this group has an adequate sanitary sewage percentage above 61%, indicating a certain trend toward efficiency for municipalities with greater sanitation coverage.

When evaluating the variable related to GDP per capita, there was also no homogeneity in the group, since the results ranged from 6,219.44 BRL in the municipality of *Piripá-BA* to 116,763.71 BRL in *Sapezal-MT*. However, considering the national GDP per capita as a parameter, which corresponds to 35,161.70 BRL [52], approximately one-third of the DMUs in this group have GDP per capita above the national index, such as *Ubarama-SP* (50,833.11 BRL), and 10% have GDP per capita close to the national index or around 30,000 BRL to 35,000 BRL, as *Sede Nova-RS* (34,369.50 BRL).

To take a picture of the behavior of efficient DMUs by group, the averages of socio-economic and environmental indicators were calculated separately (in each group) and summarized in Table 5.

Although no homogeneity was identified in the profile of the 116 DMUs, the analyses considering the averages of the

Table 5 Analysis of socio-economic and environmental indicators of efficient DMUs

Group	1	2	3	4	5	Average of groups
Population range	< 10,000	10,001 to 30,000	30,001 to 100,000	100,001 to 500,000	> 500,000	
Number of efficient DMUs	30	31	26	15	14	
Average POP	5,657	21,532	63,135	305,067	2,479,283	574,935
Average GDP (R$)	27,030.82	35,585.60	26,078.06	31,221.11	32,322.08	30,448.00
Average MHDI	0.6880	0.6930	0.6929	0.7570	0.7623	0.7158
Average DD km^2	26.06	71.77	227.71	1,924.66	2,981.61	1,046.36
Average AMS (UN)	2.00	2.50	2.00	2.50	2.43	2.40
Average (%AS)	53.56	50.36	64.01	83.01	61.19	62.43

Source: Elaborated by the authors, 2021. Database: [47].

POP, Population; MHDI, municipal human development index; DD, demographic density; AMS, average monthly salary; AS, adequate sanitary sewage

indicators (GDP per capita, HDIM, AMS and %AS (in the percentage of adequate sanitation coverage)) in each group revealed low distortions. The largest variations in the averages occurred for GDP per capita, ranging from 26,078.06 BRL (group 3) to 35,585.60 BRL (group 2), and %AS, whose variation was 50.36% to 83.03%. The variation for the MHDI was low (0.68 to 0.76), as was the variation in the amount of average monthly wages paid to formal workers (2.0 to 2.5).

With the exception of the population (POP) and demographic density (DD) variables, the results of the other indicators reveal similarities in terms of the profile of the DMUs, even though they are in different groups. These results suggest a trend in efficiency in municipalities with larger populations and with greater population density.

If one considers the averages shown to present the portrait of an efficient DMU in each group, it can be said, for example, that an efficient DMU in group 1, on average, has a population of around 5,675 inhabitants, DD around 26.04 inhabitants/km^2, GDP per capita equivalent to 27,030.82 BRL, HDMI close to 0.7, adequate sanitation coverage index above 50% and monthly remuneration to formal workers of 2 minimum wages.

Furthermore, considering the averages indicated in all groups, it can be suggested that the portrait of an efficient DMU of this group has a population average around half a million inhabitants, DD equivalent to 1,046 inhabitants/km^2, HDIM above 0.7, adequate sanitation coverage index above 60%, and monthly remuneration of formal workers around 2.4 minimum wages.

Conclusion

This research makes relevant and unprecedented contributions to the field of solid urban waste research, public managers, and other actors in society. One of the main differentials concerns the sampling scope involving 940 municipalities throughout the Brazilian territory, given that the rare studies that measure the efficiency of USWM in Brazil involve a small number of municipalities, such as de Oliveira et al. [29] who measured the efficiency of only 27 cities. Another differential to be highlighted concerns the percentage distribution of municipalities in each group by performance range, allowing for the visualization of the distance of inefficient municipalities from the efficiency frontier.

Most two-stage investigations focusing on USWM efficiency discuss the effect of socio-economic and environmental indicators on efficiency [35], lo Storto [31], Simões et al. [26], Benito-Lopez et al. [20], Expósito and Velasco [25], Simoes et al. [27], de Oliveira [29] among others, but none of these studies analyze the effect of legislation guidelines on the efficiency, as in this investigation. Therefore, in addition to discussing the relationship between socio-economic and environmental indicators with efficiency, another layer of analysis is included in this research, covering 6 guidelines of the national policy on solid waste (Law 12,305/2010). Hereby, it was possible to see whether the adherence of municipalities to these guidelines is related to efficiency and to what extent.

Another valuable contribution of this paper concerns the possibility of presenting the profile of an efficient municipality in each of the analyzed groups, considering the averages of socio-economic and environmental indicators.

The results achieved showed a group of municipalities (populations over 500,000 inhabitants) as those with the highest percentage of USWM efficiency, a result that can be explained by the possible higher collection and therefore greater availability of resources: financial, intellectual, and technological in these municipalities. This context flags up the need to implement inter-municipal actions that are capable of promoting the achievement of efficiency in less populous municipalities, through the pooling of resources.

The fact that only 12.34% of Brazilian municipalities (considering the sample) are efficient in their USWM suggests the pressing need for management remodeling, and points out that the expectations present in the NSWP are not being matched. From another perspective, the wide inefficiency evidence a reality of waste or insufficient resources used in the USWM that needs to be faced.

When considering the group of efficient municipalities, it was possible to suggest a profile among the groups having socio-economic and environmental indicators as a reference, as well as to reflect upon the effect of the NSWP guidelines on the efficiency of the USWM. The polarization found in the group of efficient municipalities with regard to adherence to the NSWP guidelines raises questions about how much the NSWP can influence the efficiency of the USWM and warns about the effectiveness of the implementation of the legislation in the municipalities.

Considering the analysis of two socio-economic and environmental indicators, the results allow us to present the profile of a DMU with a tendency toward efficiency in each analyzed group. If the averages indicated to present the picture of an efficient DMU in each group are considered, it can be said, for example, that an efficient DMU in group 1 has, on average, a population of around 5,675 inhabitants, DD around 26.04 inhabitants/km^2, GDP per capita equivalent to R$27,030.82, HDMI 0.688, percentage of adequate sanitation coverage above 50%, and monthly remuneration to formal workers of 2 minimum wages. For group 5, the portraits of the DMU are as follows: population around 2,479,283, DD 2,981.61 km^2, MHDI of 0.76, GDP per capita of 32,322.08, average monthly salary of 2.43 (salaries), and %AS is 61.19%.

Considering the defined sample, it is important to emphasize that the absence of data in the SNIS was a limiting factor when analyzing the efficiency of other USWM processes (generation, reuse, and final disposal of waste); moreover, it did not allow for comparative analyses of the performance in previous years.

The absence (in the indexed scientific bases) of similar approaches applied to Brazil was also a limiting factor, as it did not allow for comparative analyses of the results of this research with the previous investigations. The rare studies of this nature, in addition to not being indexed, do not have the sample scope proposed in this research.

The development of future research that contemplates different periods is suggested to perceive advances, stagnations, or setbacks in USWM in the country. In addition, a two-step assessment is recommended, including regression methods that identify relationships between efficiency scores and external variables.

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Declarations

Conflicts of interests The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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